1

```
C:\Users\Tharun varma>nmap -sn 192.168.1.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-12 21:12 +0530
Nmap scan report for www.domain.name (192.168.1.1)
Host is up (0.0010s latency).
MAC Address: 90:67:17:9E:EC:BB (Alphion India Private Limited)
Nmap done: 1 IP address (1 host up) scanned in 0.34 seconds
```

This scan checks whether the target host is active or not without scanning ports. It only sends ping requests and
confirms if the system is reachable

2

```
C:\Users\Tharun varma>nmap -sS scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-12 21:13 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 4.76 seconds
```

This is a SYN scan which checks the top 100 common ports. It is faster and does not complete the full TCP
connection, making it slightly stealthy. Root privileges are required to perform this scan.

3

```
C:\Users\Tharun varma>nmap -sV -p 80 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-12 21:14 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.22s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.7 ((Ubuntu))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.80 seconds
```

This scan detects the service running on port 80 and tries to identify its version. It helps in understanding what
software is being used on the server.

4

```
C:\Users\Tharun varma>nmap -O 192.168.1.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-12 21:16 +0530
Nmap scan report for www.domain.name (192.168.1.1)
Host is up (0.0043s latency).
Not shown: 995 closed tcp ports (reset)
PORT    STATE    SERVICE
21/tcp  filtered ftp
23/tcp  filtered telnet
53/tcp  open     domain
80/tcp  open     http
443/tcp open     https
MAC Address: 90:67:17:9E:EC:BB (Alphion India Private Limited)
Device type: phone
Running: Google Android 9.X|10.X|11.X, Linux 4.X
OS CPE: cpe:/o:google:android:9 cpe:/o:google:android:10 cpe:/o:google:android:11 cpe:/o:linux:linux_kernel:4
OS details: Android 9 - 11 (Linux 4.9 - 4.14)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.59 seconds
```

This scan attempts to identify the operating system of the target by analyzing network responses. It requires root
privileges for accurate detection.

5

```
C:\Users\Tharun varma\Desktop\assignment_4\Section_5>nmap -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-12 21:19 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-hostkey:
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp    open  http
|_http-title: Go ahead and ScanMe!
|_http-favicon: Nmap Project
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 14.04 seconds
```

This scan runs default NSE scripts to gather additional information such as services, configurations, and
possible vulnerabilities on the target system.